

11. Apply the principle of energy to find the horse power of an engine which can project 10000 pounds of water per min with a Velo. of 80 ft/sec

Ans. By the principle of energy

$$\text{Work done per minute} = \frac{1}{2} m v^2$$

$$= \frac{1}{2} \times 10000 \times (80)^2 \text{ foot poundals}$$

$$= \frac{10000 \times 80 \times 80}{2 \times 32} \text{ ft. lbs}$$

$$\therefore \text{So power} = \frac{10000 \times 80 \times 80}{2 \times 32 \times 60} \text{ ft. lbs/sec.}$$

1. An engine working at a constant rate H draws a load M against a resistance R ; show that the maximum speed is $\frac{H}{R}$ and the time taken to attain half this speed is $\frac{MH}{R^2} \left(\ln 2 - \frac{1}{2} \right)$. [20]

Ans. If v is the speed at any time t , the equation of motion is $M \frac{dv}{dt} = F - R$ — (1)

where M is the mass and F the force.

Since H is constant.

$$H = Fv$$

$$\text{or } F = \frac{H}{v}$$

$$\Rightarrow M \frac{dv}{dt} = \frac{H}{v} - R \quad \text{--- (2)}$$

When the speed is maximum, $\frac{dv}{dt} = 0$

$$\text{i.e. } \frac{H}{v} - R = 0$$

$$\text{or } v_{\max} = \frac{H}{R}$$

Again from (2), we have

$$M \frac{dv}{dt} = \frac{H - vR}{v}$$

$$\text{or } dt = \frac{Mv dv}{H - vR}$$

If T be the time to attain the speed $\frac{H}{2R}$.

$$\therefore \int_0^T dt = \int_0^{\frac{H}{2R}} \frac{Mv dv}{H - vR} = \frac{M}{R} \int_0^{\frac{H}{2R}} \left\{ \frac{H}{H - vR} - 1 \right\} dv$$

$$\text{or } T = -\frac{MH}{R^2} \left[\log(H - vR) \right]_0^{\frac{H}{2R}} - \frac{M}{R} \left[v \right]_0^{\frac{H}{2R}}$$

$$= -\frac{MH}{R^2} \left[\log \frac{H}{2} - \log H \right] - \frac{MH}{2R^2}$$

$$= -\frac{MH}{R^2} \left[\log \frac{1}{2} + \frac{1}{2} \right]$$

$$\text{or } T = \frac{MH}{R^2} \left[\log 2 - \frac{1}{2} \right] \quad \text{[Proved]}$$

3 Problem:- An engine works at a constant power H units of work per second. It pulls a train of total mass M against a constant resistance R . Show that the train acquires Velocity V in time $\left(\frac{MH}{R^2} \log \frac{H}{H-RV} - \frac{MV}{R} \right)$ Seconds. 20

Ans. At any instant t , let P be the force exerted by the engine and v be the Velocity of the train.

Then rate of doing work is $Pv = H$ (given),
i.e. $P = \frac{H}{v}$.

The resistance being R , the resultant force acting on the engine is $P - R = \frac{H}{v} - R$

Thus equation of motion is

$$M \frac{dv}{dt} = \frac{H}{v} - R = \frac{H - Rv}{v}$$

$$\text{or } dt = \frac{Mv dv}{H - Rv}$$

If T is the time required to acquire a Velocity V

$$\int_0^T dt = \int_0^V \frac{Mv dv}{H - Rv}$$

$$\text{or } T = \frac{M}{R} \int_0^V \frac{H - (H - Rv)}{H - Rv} dv$$

$$= \frac{M}{R} \int_0^V \left[\frac{H}{H - Rv} - 1 \right] dv$$

$$= \frac{M}{R} \left[\frac{H}{R} \log \frac{H}{H - Rv} - v \right]_0^V$$

$$T = \frac{MH}{R^2} \log \frac{H}{H - RV} - \frac{MV}{R}$$